

SOFTWARE REQUIREMENTS SPECIFICATION

School Website Management System

Prepared By	Esha Zafar
Roll No.	4064
Submitted To	Abdul Karim Nawaz
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Institute of Computing & Information Technology

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the School Website Management System (SWMS). The purpose of this document is to present a detailed description of the system's functionalities, constraints, interfaces, and interactions with the external world. It is intended for developers, project managers, testers, and stakeholders involved in the design, development, deployment, and maintenance of the system.

The SWMS aims to digitize and centralize school operations, making administration efficient, communication seamless, and information easily accessible to all stakeholders including students, parents, teachers, and administrative staff.

1.2 Scope

The School Website Management System is a web-based application designed to manage all core activities of a school. The system encompasses the following high-level features:

- Online student registration and admission management
- Academic calendar management and scheduling
- Result and grade management with automatic report card generation
- Staff and teacher profile management
- Fee management, invoicing, and payment tracking
- News, notices, and event publishing portal
- Online library catalog and resource management
- Attendance tracking for students and staff
- Parent and guardian communication portal
- Examination timetable and hall ticket generation

The system is intended to replace manual processes with automated digital workflows, reducing paperwork, minimizing errors, and enabling real-time access to critical school information.

1.3 Definitions, Acronyms & Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
SWMS	School Website Management System
Admin	Administrator — person responsible for managing the entire system
UI	User Interface — the front-end visual layer presented to users

DB	Database — structured storage system for persistent data
DFD	Data Flow Diagram — visual representation of data flow in the system
ER Diagram	Entity Relationship Diagram — shows relationships between data entities
UML	Unified Modeling Language — standard modeling language for software design
CRUD	Create, Read, Update, Delete — basic database operations
HTTP/HTTPS	HyperText Transfer Protocol (Secure) — web communication protocol
CMS	Content Management System — system to manage digital content
API	Application Programming Interface — interface for software components to communicate

1.4 Overview

The remainder of this SRS document is structured as follows: Section 2 provides an overall description of the system including product perspective, features, and user characteristics. Section 3 details all functional requirements. Section 4 covers non-functional requirements. Sections 5–8 present system diagrams including Use Case, DFD, ER, and Architecture diagrams. Sections 9–10 describe external interfaces and the appendix.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The School Website Management System is a standalone, web-based application that serves as the central digital hub for all school activities. It is designed to replace fragmented manual processes with an integrated system that maintains all data within a centralized database. The system interacts with users through a responsive web interface accessible on desktops, tablets, and mobile phones.

It is not dependent on any existing legacy system but may integrate with external systems such as SMS gateways for notifications, email servers for communication, and payment gateways for online fee collection. The system uses a client-server architecture where the server hosts both the application logic and the database.

2.2 Product Functions

The major functions of the SWMS include:

#	Function	Description
1	User Authentication	Secure login/logout with role-based access control for Admin, Teachers, Students, and Parents
2	Student Management	Enroll, update, and manage student records, profiles, and academic history
3	Teacher Management	Manage teacher profiles, assign classes, and track performance
4	Class & Schedule Mgmt	Create and manage class timetables, room allocations, and academic periods
5	Attendance System	Daily attendance marking with automated reports and absence notifications
6	Examination Module	Manage exam schedules, hall tickets, result entry, and grade computation
7	Fee Management	Track fee structures, generate invoices, record payments, send reminders
8	Library Management	Book catalog, issue/return tracking, and availability search
9	News & Events	Publish school news, upcoming events, holidays, and circulars
10	Reports & Analytics	Generate academic, financial, and attendance reports in PDF/Excel

2.3 User Classes and Characteristics

User Class	Description	Access Level
Super Admin	System administrator with full control over all modules and settings	Full Access
School Admin	Manages day-to-day operations, admissions, fees, and reports	High Access
Teacher	Manages classes, marks attendance, enters grades, views schedules	Medium Access
Student	Views own results, timetable, notices, fee status, and library books	Limited Access

Parent	Views child's progress, attendance, fees, and communicates with school	Read Only
Librarian	Manages book catalog, issues, returns, and library reports	Module-Specific

2.4 Operating Environment

The School Website Management System shall operate in the following environment:

- Server OS: Linux (Ubuntu 20.04 LTS or higher) or Windows Server 2019+
- Web Server: Apache 2.4+ or Nginx 1.18+
- Backend Language: PHP 8.0+ / Python 3.9+ / Node.js 16+
- Database: MySQL 8.0+ or PostgreSQL 14+
- Frontend: HTML5, CSS3, JavaScript (ES6+), Bootstrap 5
- Browser Compatibility: Google Chrome 90+, Mozilla Firefox 88+, Microsoft Edge 90+, Safari 14+
- Mobile Compatibility: Fully responsive design supporting Android 8+ and iOS 13+
- Minimum Server RAM: 4 GB (8 GB recommended for production)
- Internet Connectivity: Required for cloud-based deployment; LAN support for on-premises

2.5 Assumptions and Dependencies

The following assumptions and dependencies apply to this system:

- It is assumed that the school has a stable internet connection for cloud-based deployment.
- All users are assumed to have basic computer literacy and access to a modern web browser.
- The system depends on a valid SSL certificate for secure HTTPS communication.
- Email and SMS notification features depend on third-party SMTP and SMS gateway services.
- Payment integration depends on availability of a supported payment gateway (e.g., JazzCash, EasyPaisa, or Stripe).
- The system assumes that user data entry is accurate; the system is not responsible for incorrect manually entered data.
- Regular database backups are the responsibility of the system administrator.

3. SYSTEM FEATURES & FUNCTIONAL REQUIREMENTS

3.1 Admin Module

The Admin Module is the core management hub of the SWMS. The administrator has privileged access to all system components.

3.1.1 User Management

The system shall allow the administrator to:

- Create, edit, deactivate, and delete user accounts for all roles
- Assign and modify role-based permissions
- Reset user passwords and manage session controls
- View activity logs and audit trails for all users

3.1.2 Academic Year Management

The system shall enable the administrator to:

- Define and manage academic sessions, terms, and semesters
- Set term start/end dates and configure holiday calendars
- Promote students from one class to the next at end of academic year
- Archive historical academic year data for record-keeping

3.1.3 Admission Management

The admin shall be able to:

- Configure and publish online admission forms with customizable fields
- Review, approve, or reject student applications with comments
- Generate admission confirmation letters and fee challans
- Set admission criteria, class capacity, and seat availability

3.2 Student Module

The Student Module provides students with access to their academic information and school resources.

- Register and maintain personal profile with photo upload
- View class timetable and academic calendar
- Check attendance records and download attendance reports
- View published results, grade reports, and performance analytics
- Download fee invoices, view payment history, and pay fees online
- Search library catalog and view issued books with due dates
- Access study material, assignments, and downloadable resources

- View school news, upcoming events, and official circulars
- Submit feedback and queries through the student portal

3.3 Teacher Module

The Teacher Module provides teachers with tools to manage their classes and student performance.

- View assigned classes, subjects, and student rosters
- Mark daily attendance for each class and submit attendance records
- Enter examination marks, assignment grades, and comments
- Generate and download class-wise result reports
- Upload study material, assignments, and lecture notes for students
- Communicate with students and parents through the messaging portal
- View personal timetable, leave records, and announcements
- Submit leave applications and view leave approval status

3.4 News & Events Module

This module manages all public-facing content on the school website.

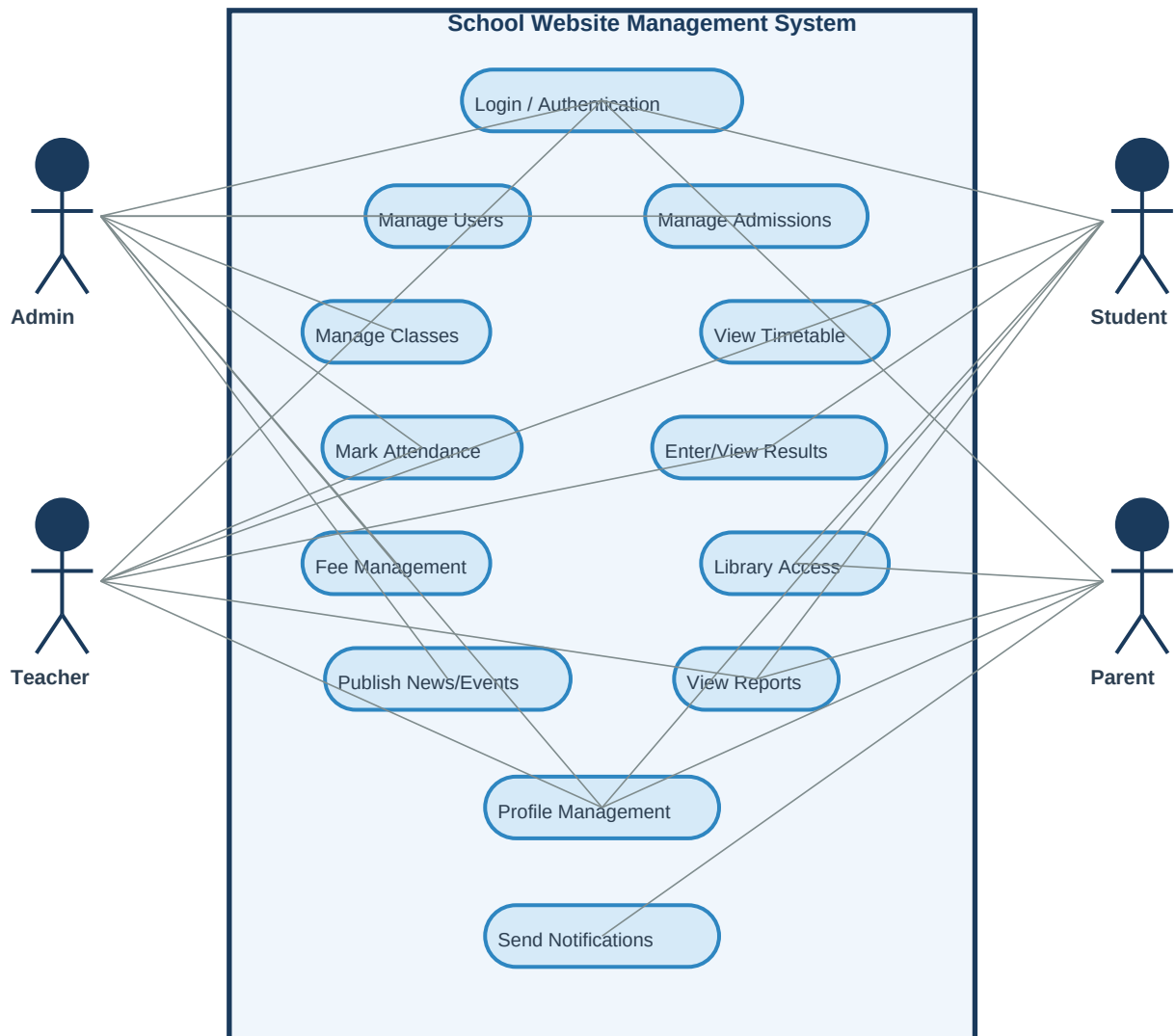
- Admin can create, edit, publish, and archive news articles with images
- Events can be added to the school calendar with date, venue, and description
- Gallery management for uploading and organizing school event photos
- Notice board for publishing urgent announcements with priority levels
- Automated email/SMS notification to users when new content is published
- Public-facing homepage displays latest news, upcoming events, and notices

4. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement	Metric / Standard
Performance	System response time for standard page load	< 3 seconds under normal load
Performance	System must handle concurrent users	Minimum 200 simultaneous users
Scalability	Database must support large datasets	Up to 10,000 student records
Security	All passwords must be encrypted	bcrypt hashing (min cost 10)
Security	Data transmission must be encrypted	TLS 1.2 / SSL encryption
Security	Protection against common web attacks	XSS, SQL Injection, CSRF prevention
Availability	System uptime guarantee	99.5% uptime (max 3.65 hrs/year downtime)
Usability	Intuitive UI requiring minimal training	New user productive within 2 hours
Usability	Fully responsive on all devices	Mobile, tablet, desktop compatibility
Reliability	Automatic database backups	Daily incremental, weekly full backup
Maintainability	Modular code architecture	MVC pattern, documented codebase
Portability	Cross-browser compatibility	Chrome, Firefox, Edge, Safari support
Compliance	Data privacy and protection	Compliant with local data protection laws

5. USE CASE DIAGRAM

The Use Case Diagram below illustrates all system actors and their interactions with the School Website Management System. Four primary actors are identified: Administrator, Teacher, Student, and Parent.

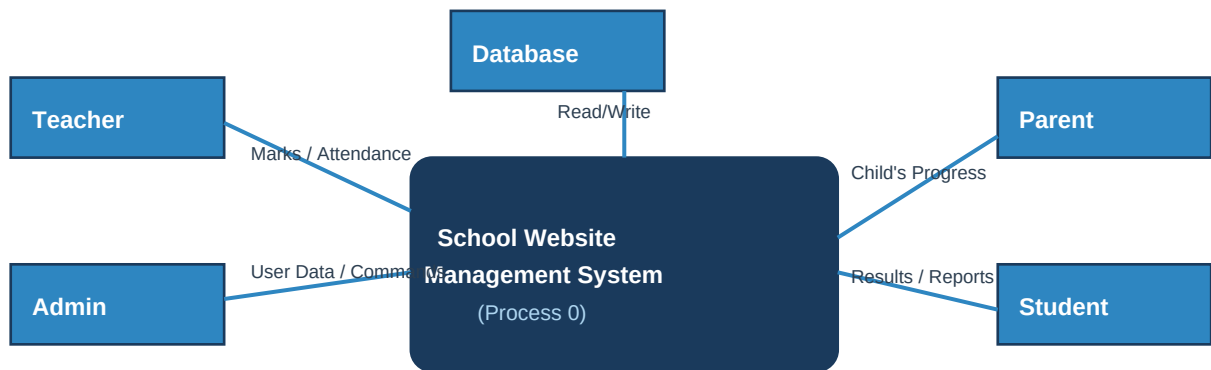


The diagram shows how each actor interacts with the system's core functionalities. The Administrator has the broadest set of interactions, while Parents primarily have read-only access to view their child's information and communicate with the school.

6. DATA FLOW DIAGRAM (DFD) — Level 0 & Level 1

6.1 Level 0 DFD (Context Diagram)

The Level 0 DFD shows the School Website Management System as a single process with all external entities and data flows.



6.2 Level 1 DFD (Expanded Processes)

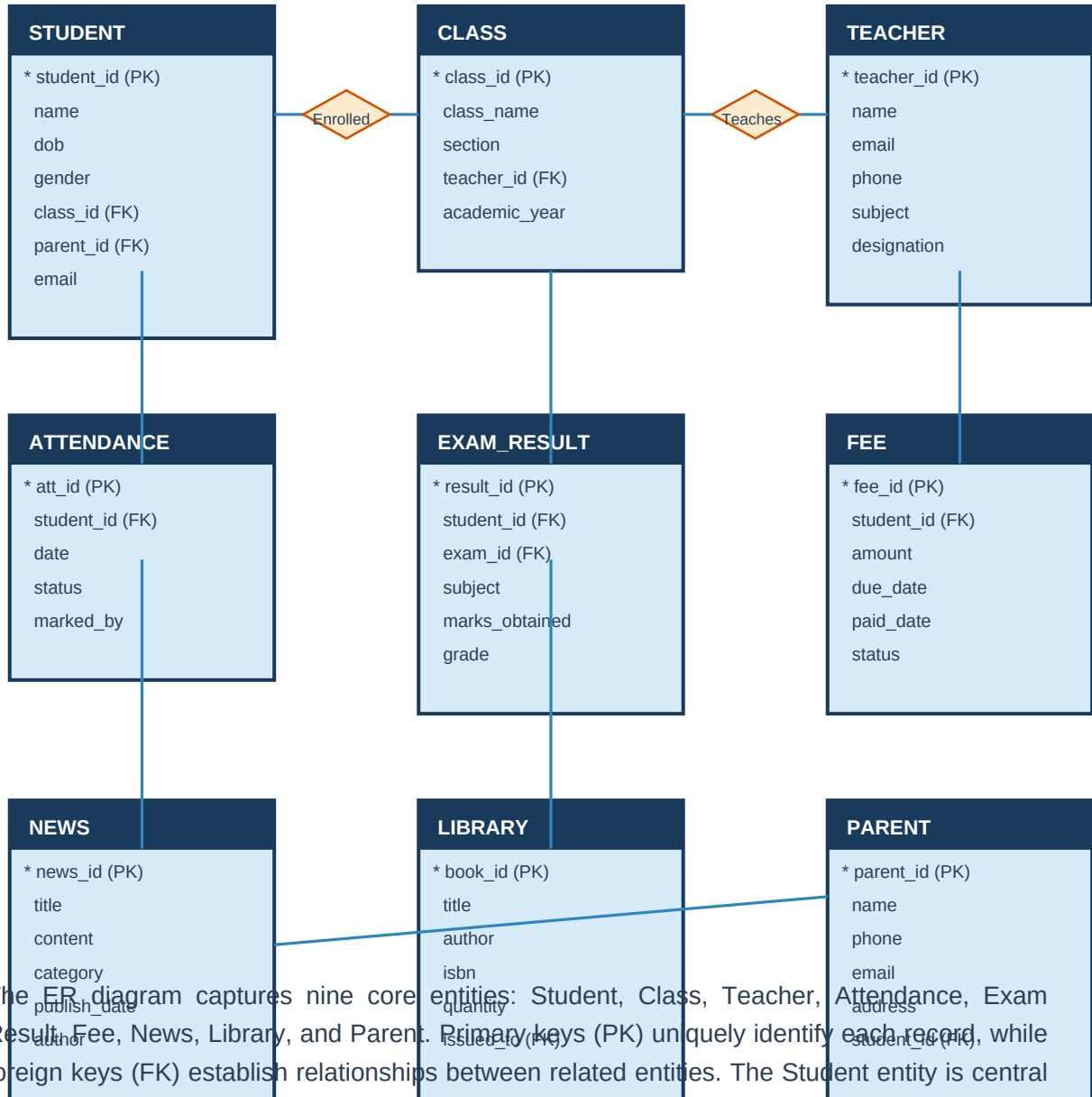
The Level 1 DFD expands the central system into key sub-processes, showing how data flows between them and the external entities.



Each process in the Level 1 DFD handles a specific domain of the system. Data flows from external actors through authentication into the various functional modules, which in turn read and write to the persistent database stores.

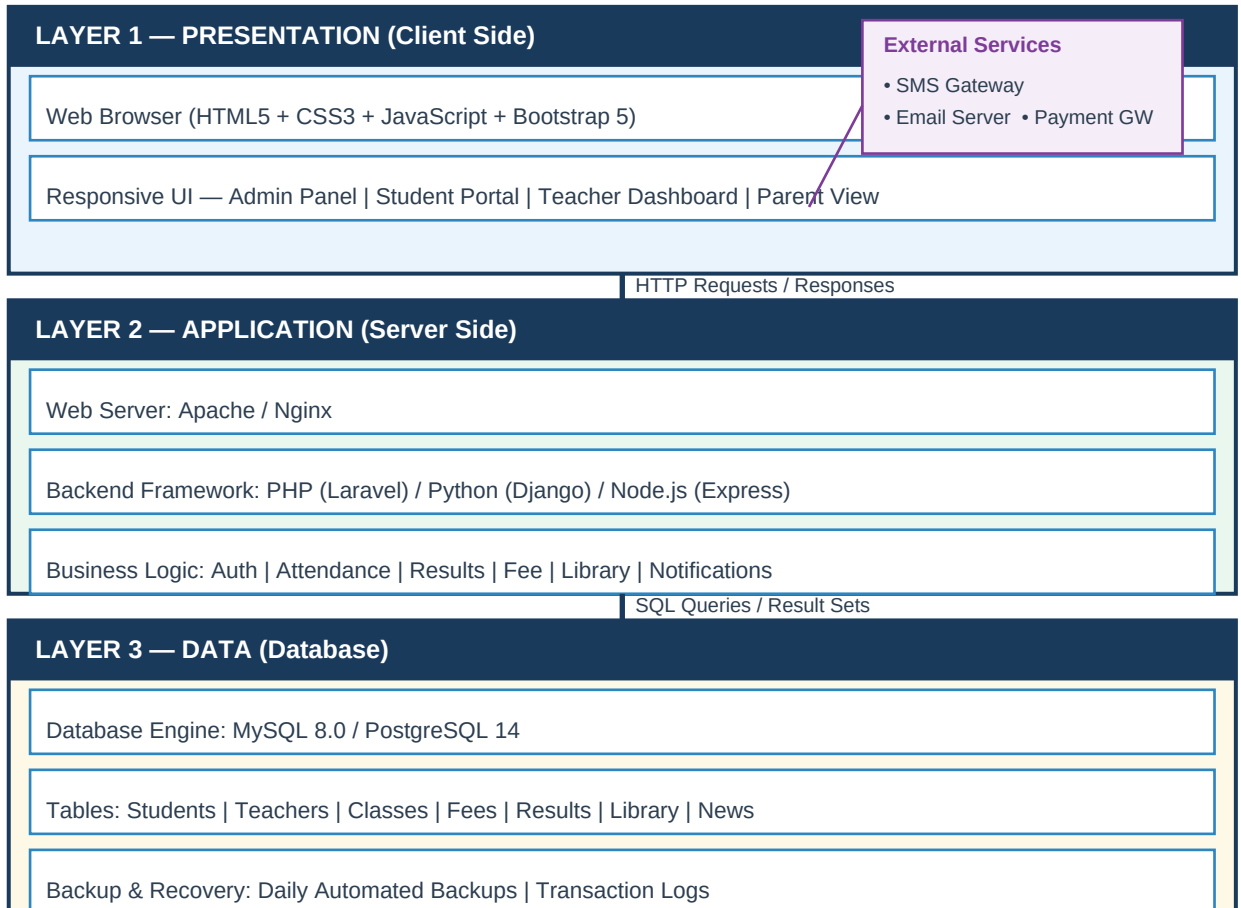
7. ENTITY RELATIONSHIP (ER) DIAGRAM

The ER diagram illustrates the data model of the School Website Management System, showing all major entities, their attributes, and the relationships between them.



8. SYSTEM ARCHITECTURE DIAGRAM

The system follows a three-tier architecture comprising the Presentation Layer (Client), Application Layer (Server), and Data Layer (Database). This architecture ensures separation of concerns, scalability, and maintainability.



The three-tier architecture separates the system into client, server, and database layers. This design allows each layer to be developed, tested, and scaled independently. External services such as SMS gateways, email servers, and payment gateways are integrated at the Application Layer through secure API calls.

9. EXTERNAL INTERFACE REQUIREMENTS

9.1 User Interface Requirements

- The UI shall be clean, intuitive, and consistent across all modules
- Navigation menu shall be hierarchical and accessible from all pages
- The system shall display confirmation dialogs for all destructive actions
- Form validation shall be performed both client-side (JavaScript) and server-side
- Error messages shall be descriptive and guide users toward resolution
- The dashboard shall display key KPIs using graphical charts and summaries
- The system shall support both Light Mode and Dark Mode themes

9.2 Hardware Interface Requirements

- The system shall operate on standard server hardware with minimum 4-core CPU
- Client side requires only a web-enabled device with keyboard/mouse or touchscreen
- Network interface: 100Mbps LAN or higher; 10Mbps internet connection minimum
- Storage: Minimum 50 GB server storage for application, media, and database

9.3 Software Interface Requirements

Interface	Type	Purpose
MySQL / PostgreSQL	Database	Primary persistent data storage
SMTP Server (Gmail/SendGrid)	Email Gateway	User registration, notifications, alerts
SMS Gateway (Twilio/Jazz)	SMS API	Attendance alerts, fee reminders, results
Payment Gateway (Stripe/EasyPay)	Payment API	Online fee collection
AWS S3 / Local Storage	File Storage	Document uploads, images, reports
Google Maps API	Maps API	School location display on contact page
PDF Generator (FPDF/wkhtmltopdf)	Library	Report cards, fee invoices, hall tickets

9.4 Communication Interface Requirements

- All data transmission shall use HTTPS with TLS 1.2 or higher encryption
- REST API architecture shall be used for backend-frontend communication
- JSON format shall be used for all API request and response payloads
- WebSocket support shall be implemented for real-time notifications and chat
- Session tokens shall expire after 30 minutes of inactivity for security

10. APPENDIX

10.1 Version History

Version	Date	Author	Description
0.1	15 April 2026	Esha Zafar	Initial draft — Introduction & Scope
0.5	22 April 2026	Esha Zafar	Added functional requirements and user classes
0.8	28 April 2026	Esha Zafar	Added diagrams, non-functional requirements
1.0	01 May 2026	Esha Zafar	Final version submitted to Abdul Karim Nawaz

10.2 References

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- [2] Sommerville, I. (2016). Software Engineering (10th Edition). Pearson Education Limited.
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- [4] OWASP Foundation (2023). OWASP Top Ten Web Application Security Risks.
- [5] W3C Web Content Accessibility Guidelines (WCAG) 2.1
- [6] Bootstrap Documentation v5.3 — <https://getbootstrap.com/docs/>
- [7] MySQL 8.0 Reference Manual — <https://dev.mysql.com/doc/>

10.3 Approval Sign-Off

Role	Name	Signature	Date
Author	Esha Zafar (Roll No. 4064)	_____	01 May 2026
Instructor	Abdul Karim Nawaz	_____	_____
HOD / Reviewer	_____	_____	_____

This document is prepared as part of a Software Engineering course assignment. All requirements described herein are based on analysis of school management needs and standard SRS practices as defined by IEEE Std 830-1998.